Pearls AQI Predictor

Air quality forecasting with machine learning and deep learning — delivering production-ready predictions up to 72 hours ahead.

By Mansoor Abid

What the System Does

Fully Automated

Serverless ML/DL pipeline forecasting AQI up to 72 hours ahead — no manual intervention required.

Auto-Promotion

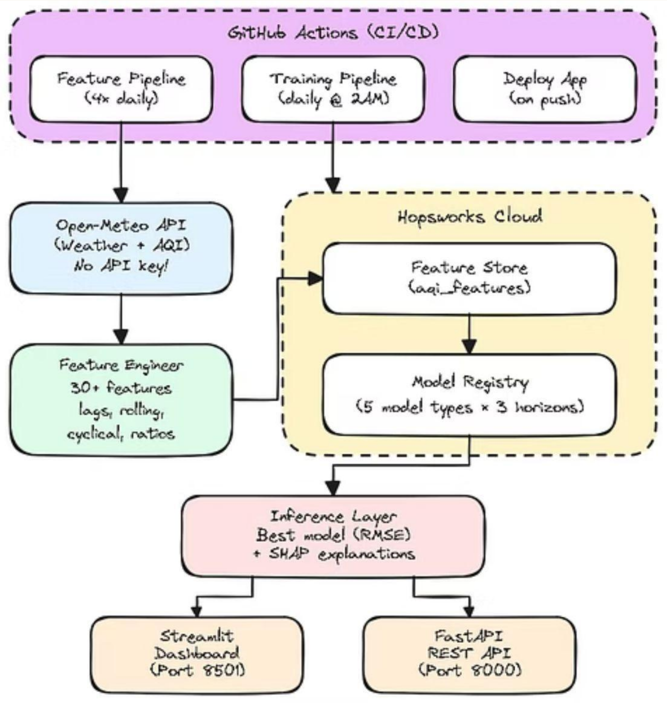
Evaluates 5 architectures daily and promotes the best performer to production based on skill_vs_naive and RMSE.

Live Dashboard

Real-time interactive dashboard with explainable predictions and health status indicators.

Made with GAMMA

System Architecture



Data Layer

Hourly weather & pollutant data ingested from the Open-Meteo API — no API key required.

Pipeline

Runs 4× daily via GitHub Actions, pushing features into the Hopsworks Feature Store for versioned, reproducible training.

Model Zoo

Five architectures compete every cycle: **Ridge, Random Forest, XGBoost, Neural Networks,** and **LSTM**.

Made with GAMMA

Feature Engineering

30+ engineered signals give every model a rich, domain-aware view of the atmosphere:

Temporal Cycles

Sine/cosine encodings capturing diurnal and seasonal patterns.

Lag Features

1h, 3h, 6h, and 24h lags preserving recent pollutant memory.

Rolling Statistics

Moving averages and standard deviations over multiple windows.

Domain Ratios & Wind

$PM_{2.5}/PM_{10}$, NO_2/O_3 ratios plus wind vector decomposition for dispersion modeling.

Made with GAMMA

Model Performance

0.48

Skill Score

48% better accuracy than the naive baseline, but it is changing continuously as the data accumulates.

~9.26

RMSE (AQI pts)

Root mean squared error for the best-performing 1-day horizon model. Initial experiments: ~18 RMSE → Current production: 9.26 RMSE

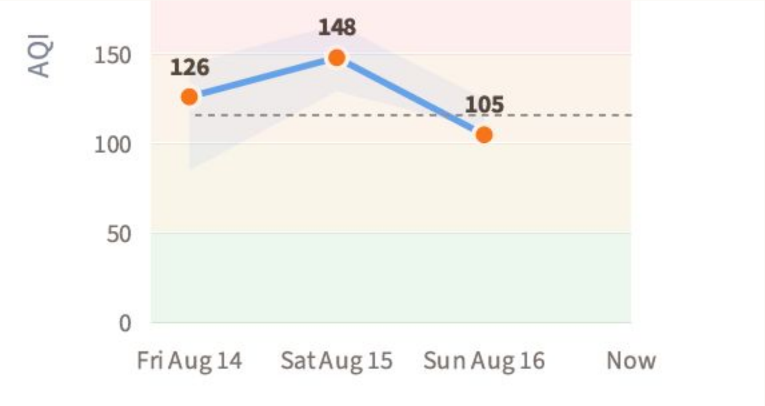
Best Model: Random Forest

- Across daily competitions, Random Forest consistently wins on the **1-day horizon**; balancing bias, variance, and interpretability.
- The skill score of **0.48** means the system is nearly twice as informative as a simple persistence forecast; the standard for operational AQI tools.

Made with GAMMA

Model Performance

Day	Model	Version	RMSE
Day 1	random_forest	7	9.26
Day 2	random_forest	3	9.49
Day 3	random_forest	3	10.17



Made with GAMMA

Three Prediction Horizons

1

2

3

Day 1

Next 24 hours — highest accuracy, ideal for immediate health decisions and public alerts.

Day 2

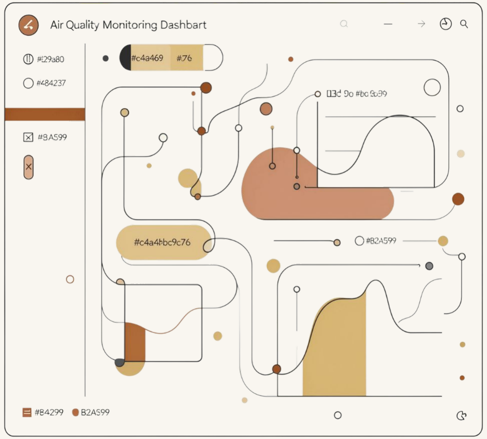
24–48 hours ahead — reliable mid-range outlook for planning outdoor activities.

Day 3

48–72 hours ahead — strategic foresight for event scheduling and policy responses.

Made with GAMMA

Dashboard Features



Real-Time AQI Gauge

Live health status indicators color-coded to WHO thresholds — instantly readable at a glance.

Pollutant Breakdown

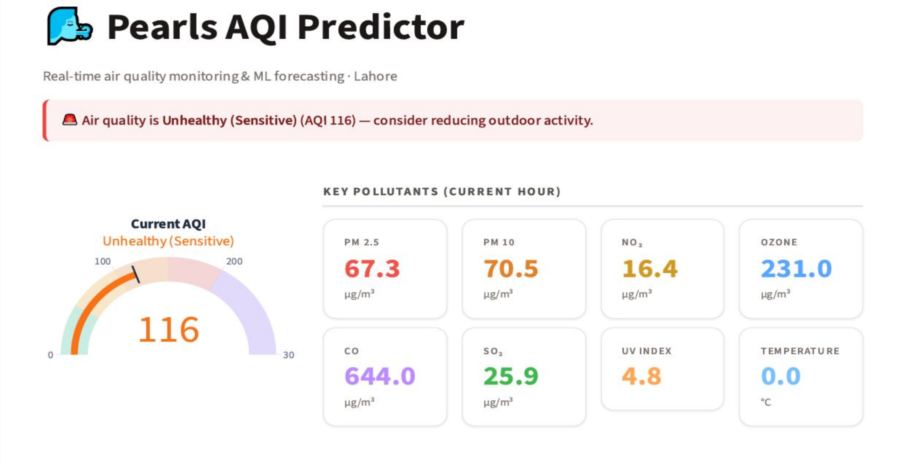
Detailed view of PM2.5, NO_2, O_3, and other key constituents driving the current index.

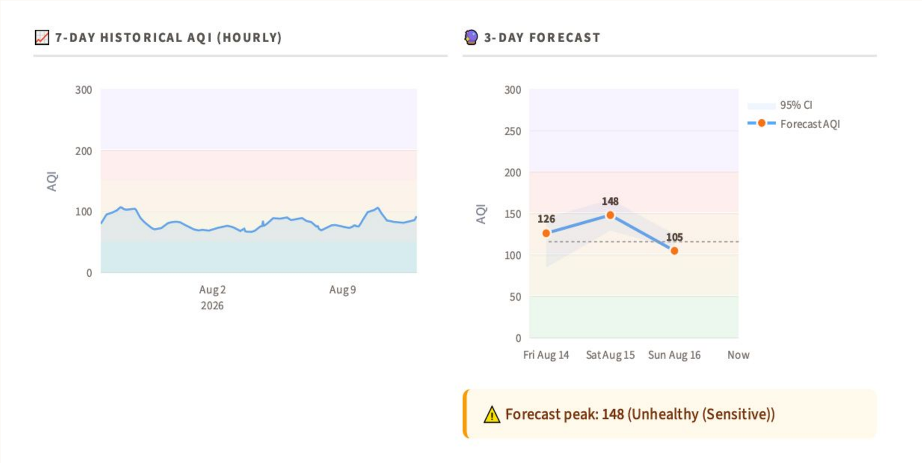
History + Forecast

7-day historical chart alongside a 3-day forecast with confidence bands for uncertainty communication.

Made with GAMMA

Dash board





Made with GAMMA

Pearls AQI

City: Lahore

Updated: 11:33, 13 Aug 2026

Refresh data

Forecast Horizon

1 2 3

AQI GUIDE

0-50	Good
0-100	Moderate
0-150	Unhealthy (Sensitive)
0-200	Unhealthy
0-300	Very Unhealthy
0-500	Hazardous

ACTIVE MODELS

Day	Model	Version	RMSE
Day 1	random_forest	7	9.26
Day 2	random_forest	3	9.49
Day 3	random_forest	3	10.17

WEATHER SNAPSHOT

Humidity —

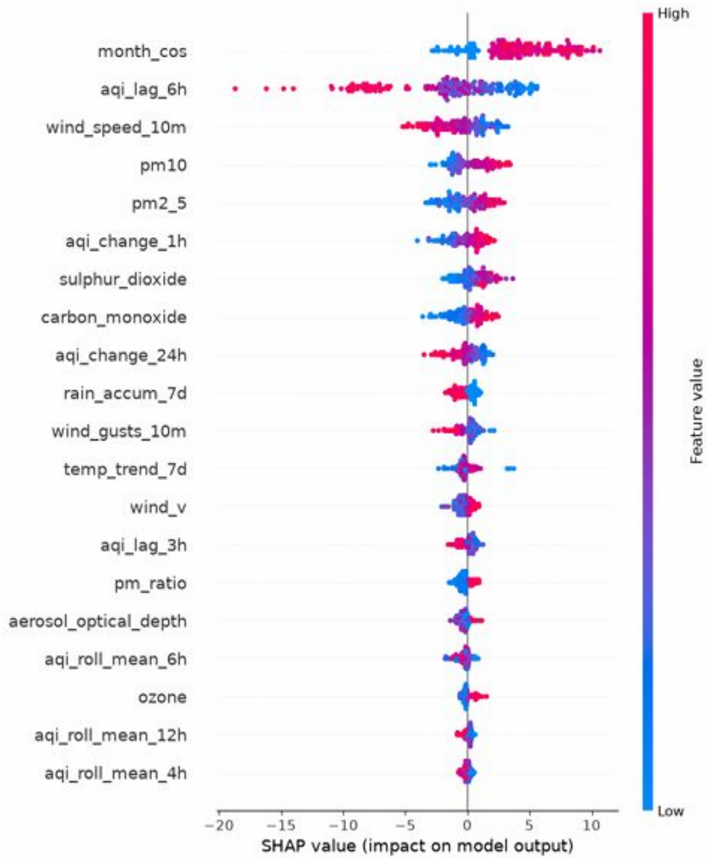
Wind —

Rain —

Cloud cover —

FEATURE IMPORTANCE (SHAP)

Model: random_forest · Horizon: 1d



GAMMA

Explainability with SHAP

Every prediction is accompanied by a transparent, auditable explanation — not just a number.

Feature Importance Breakdown

SHAP values decompose each forecast to show exactly which signals drove the result.

Top Drivers

month_cos, aqi_1h_lag, aqi_6h_lag, and wind_speed_10m dominate predictions.

Note: *Feature rankings evolve with retraining — these reflect the current production model.*

Audit Trail

All SHAP visualizations are versioned and saved to Hopsworks for compliance and review.

Made with GAMMA

Quick Start

Requirements

- Python 3.12+
- Hopsworks account (free tier)
- Open-Meteo API — no key needed

1. Clone the repo
2. Run `uv sync`
3. Fill `.env` credentials
4. Launch the dashboard

Made with GAMMA

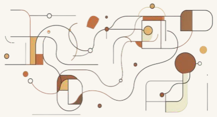
Future Works

Multi-City Expansion

Extend coverage to Karachi, Islamabad, and other major urban centers with high pollution burden.

Drift Triggered Retraining

Implement automated data drift monitoring to trigger model retraining when distribution shifts are detected.



Ensemble Models

Blend the top 3 performing models into a single ensemble for improved robustness and lower error variance.

Note: All the data and metrics are subject to change because of continuous data accumulation and retraining.

Made with GAMMA